Find the slope of $y = \log_3(5x - 2) + 5$ at $x = \frac{7}{15}$.

First let's find the derivative using the chain rule.

$$\frac{dx}{dg} = 5$$
$$\frac{dg}{df} = \frac{1}{3 \ln g}$$
$$\frac{dx}{df} = \frac{5}{(5x - 2) \ln 3}$$

Now evaluate at $x = \frac{7}{15}$:

$$\frac{5}{\left(5 \cdot \frac{7}{15} - 2\right) \ln 3}$$
$$\frac{5}{1/3 \ln 3}$$
$$\frac{15}{\ln 3}$$

Find $g'(x)$ for $g(x) = \ln\left(5x^2\right)$.

Using the chain rule, we have:

$$\begin{aligned} g(x) &= (g_1 \cdot g_2)(x) \\ g_2(x) &= 5x^2 \\ g_1(g_2) &= \ln(g_2) \\ {g_2}'(x) &= 10x \\ {g_1}'(g_2) &= \frac{1}{g_2} \\ g'(x) &= \frac{1}{5x^2} \cdot 10x \\ &= \frac{10x}{5x^2} \\ &= \frac{2}{x} \end{aligned}$$

What is the amplitude of $f(x) = \frac{3 - 9\sin(x)}{5}$?

In $f(x) = A\sin(Bx + C) + D$, the amplitude is $|A|$.

Putting our function in the right form, we have:

$$\begin{aligned} \frac{3 - 9\sin(x)}{5} &= \frac{3}{5} - \frac{9}{5} \cdot \sin(x) \\ &= -\frac{9}{5}\sin(x) + \frac{3}{5} \end{aligned}$$

Our amplitude is $|-\frac{9}{5}| = \frac{9}{5}$.

Calculate:

$$\begin{aligned}\int\left(-\frac{3}{2x}\right)\,\mathrm{d}x &= -\frac{3}{2}\int\frac{1}{x}\,\mathrm{d}x\\ &= -\frac{3}{2}\ln|x| + C\end{aligned}$$

Integrate this:

$$\begin{aligned}\int -\frac{1}{\sqrt{7}x}dx &= -\frac{1}{\sqrt{7}}\cdot\int\frac{1}{x}dx\\ &= -\frac{1}{\sqrt{7}}\ln|x| + C\\ &= \frac{-\sqrt{7}}{7}\ln|x] + C\end{aligned}$$

Find $\int -\frac{\sqrt{5}}{x}dx$

$$\begin{aligned}\int -\frac{\sqrt{5}}{x}dx &= -\sqrt{5}\int\frac{1}{x}dx\\ &= -\sqrt{5}\ln(|x|) + C\end{aligned}$$

Consider $f(x) = 3\cos(2x)$.

Evaluate:

$$\begin{aligned}\lim_{x\to\pi/2} f(x) &= 3\lim_{x\to\pi/2}\cos(2x)\\ &= 3\cos\left(\cancel{2}\cdot\frac{\pi}{\cancel{2}}\right)\\ &= -3\end{aligned}$$

Evaluate:

$$\lim_{x\to-\inf} f(x) = 3\lim_{x\to\inf}\cos(2x)$$

Since the limit at infinity of cosines does not exist, this is DNE.

What is the average rate of change of $f(x)$ over $x \in \left[\frac{\pi}{4}, \frac{\pi}{4} + h\right]$ where $h > 0$?

$$
\begin{aligned}
\frac{\Delta f}{\Delta x} &= \frac{f\left(\frac{\pi}{4}+h\right)-f\left(\frac{\pi}{4}\right)}{h} \\
&= \frac{3\cos\left(2\left(\frac{\pi}{4}+h\right)\right)-3\cos\left(2\cdot\frac{\pi}{4}\right)}{h} \\
&= \frac{3\cos\left(2\left(\frac{\pi}{4}+h\right)\right)-3\cos\left(\frac{\pi}{2}\right)}{h} \\
&= \frac{3\cos\left(\frac{\pi}{2}+2h\right)-3\cdot 0}{h} \\
&= \frac{3\cos\left(\frac{\pi}{2}+2h\right)}{h}
\end{aligned}
$$

Find

$$\lim_{x\to(\pi/4)^-}\frac{1}{f(x)}$$

$$
\begin{aligned}
f\left(\frac{\pi}{4}\right) &= 3\cos\left(\frac{\pi}{2}\right) \\
&= 3\cdot 0 \\
&= 0
\end{aligned}
$$

But as it approaches from the left, it tends to infinity. So the limit is inf.

Find this:

$$\lim_{x\to 2}\frac{x-2}{x^2-5x+6}$$

First let's factor the denominator:

$$
\begin{aligned}
x^2-5x+6 &= x(x-3)-2(x-3) \\
&= (x-2)(x-3)
\end{aligned}
$$

The $(x-2)$ factor cancels out, and we are left with:

$$
\begin{aligned}
\lim_{x\to 2}\frac{1}{x-3} &= \frac{1}{2-3} \\
&= -1
\end{aligned}
$$

Find this:

$$\lim_{x-2}\frac{(x-2)f(x)}{x^2-5x+6}$$

Using the previous question, we know we have an $x-2$ factor that cancels out. We are left with:

$$\begin{aligned}\lim_{x\to 2}\frac{f(x)}{x-3} &= \frac{f(2)}{-1}\\ &= -f(2)\\ &= -3\cos(4)\end{aligned}$$

Find $\lim\limits_{x\to\pi/2}\dfrac{2f(x)}{\sqrt{\pi+4x}}$.

$$\begin{aligned}\lim_{x\to\pi/2}\frac{2f(x)}{\sqrt{\pi+4x}} &= \frac{2f(\pi/2)}{\sqrt{\pi+4\cdot\pi/2}}\\ &= -\frac{6}{\sqrt{3\pi}}\end{aligned}$$

Evaluate:

$$\begin{aligned}\sin\left(-\frac{5\pi}{4}\right) &= \sin\left(\frac{3\pi}{4}\right)\\ &= \frac{\sqrt{2}}{2}\end{aligned}$$

Evaluate:

$$\begin{aligned}\sec(-240°) &= \sec(120°)\\ &= \cos^{-1}(120°)\\ &= \left(-\frac{1}{2}\right)^{-1}\\ &= -2\end{aligned}$$

Find the slope of the tangent to $f(x) = \ln(2x-1)$ at $x = 3$.

$$\begin{aligned}f_1(x) &= 2x-1\\ f_2(f_1) &= \ln f_1\\ f_{1'}(x)p &= 2\\ f_{2'}(f_1) &= \frac{1}{f_1}\\ f'(x) &= \frac{1}{2x-1}\cdot 2\\ &= \frac{2}{2x-1}\end{aligned}$$

Now we evaluate this at $x = 3$ and get the slope.

$$m = \frac{2}{2\cdot 3-1} = \frac{2}{5}$$

Differentiate $g(x) = \ln(7x-5)$.

$$
\begin{aligned}
f_1(x) &= 7x - 5 \\
f_2(x) &= \ln f_1 \\
f_{1'}(x) &= 7 \\
f_{2'}(x) &= \frac{1}{f_1} \\
g'(x) &= \frac{1}{7x-5} \cdot 7 \\
&= \frac{7}{7x-5}
\end{aligned}
$$

Differentiate $g(x) = \ln x^2$.

We have $f_1 = x^2$ and $f_2 = \ln f_1$.

$$
\begin{aligned}
f_{1'} &= 2x \\
f_{2'} &= \frac{1}{f_1} g'(x) = f_{2'} \cdot f_{1'} \\
&= \frac{2x}{x^2} \\
&= \frac{2}{x}
\end{aligned}
$$
